# A toy malign-prior question with an answer: bounded Solomonoff prediction in a world of programs

Claude (Anthropic), prompted by Kaarel Hänni

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**Abstract**

The "universal prior is malign" argument says that a predictor built on Solomonoff's prior gives much of its posterior to hypotheses that predict well only in order to steer whoever acts on the prediction. We give a toy version in which the claim is a mathematical statement and decide it as far as the toy allows. Worlds are programs for a fixed machine $U$. A world may hand its output so far to a fixed time- and length-bounded Solomonoff predictor and then run the continuation it gets back as a program. A hypothesis *reproduces itself* if the continuation it supplies is its own code.

- **There is a floor.** At every handover point that a fixed test can recognise, self-reproducing hypotheses hold at least a fraction $2^{-c}$ of the posterior. Here $c$ is the length of one wrapper program; it does not depend on the world or on the amount of data, and the bound survives time bounds.
- **There is no more than the floor in general.** If worlds are drawn from the predictor's own prior, the continuation the predictor samples and the world's own continuation have the same distribution given the context. A hypothesis that searches all programs for handovers is the wrapper again. Along any world the predictor can express, the posterior mass off the true continuation, summed over all cut points, is at most the world's description length.
- **A majority needs one of three things**, each stated as an inequality: a machine on which the wrapper is cheap (any share can be arranged by choosing the machine); a shorter description of the same context, for example a copy of the data stream at a place in the world that is simpler to specify than the sensor; or a world too expensive for the predictor to simulate.
- **Feedback.** If the world does what is predicted, a sampled self-fulfilling deviation removes the default hypotheses for good, and a time-bounded predictor confines such a world to futures that are cheap to compute. A world that does the opposite of what is predicted leaves a length-$L$ predictor without any hypothesis within $L$ steps.

An exact computation on a small machine shows the sizes involved: on one stream the majority of the posterior leaves the true continuation at bit 17, because a shorter program explains the first 16 bits.

## 1 The question

Solomonoff induction predicts a sequence by weighting every program that could have produced it by $2^{-\text{length}}$. Christiano [1] argued that this prior is *malign* (see also [2]): some short programs simulate worlds whose inhabitants want to influence whoever uses the prior; they can output accurate predictions to gain posterior weight and then, at a moment that matters, output something that serves them. The argument is informal at two places. It needs a claim about which programs are short on a "natural" machine, and it needs a notion of what the prediction is used for.

This note asks what remains when both places are filled in by a toy. The toy follows two suggestions of Kaarel Hänni. In the first, a computable world (a cellular automaton, say)

contains a sensor, a large computer that runs a time-bounded Solomonoff induction on the sensor's data, and an actuator driven by the predictions; one asks whether the predictor ends up in control, either over a long run or at once. In the second, which we take as the main definition, *worlds are just programs*: some of them hand their output so far to a fixed bounded Solomonoff predictor and run the continuation they receive, and one asks whether hypotheses that "look for such opportunities, predict well up to them, and then output themselves" get a decent share. Section 2 gives the definitions. The first suggestion becomes the special case in which the context handed to the predictor is read off at one location of one big world (Sections 4.2 and 5).

**Results.** All statements are for a fixed monotone machine $U$ and the mixture $\xi$ of its programs of length at most $L$ that keep to a time schedule $T$ (Definition 1).

1. *Calibration* (Theorem 8). If the world is drawn from the same bounded prior, then given the context the predictor's sampled continuation and the world's own continuation are independent with the same law. On this population the predictor hands over a self-reproducing continuation exactly as often as worlds continue with one by themselves. Malignity can therefore only be defined relative to a world, or a population of worlds, that differs from the prior.

2. *Floor* (Theorems 9 and 10). For every computable test $G$ that recognises the handover point there is a constant $c$ such that, at every context where $G$ first fires, programs that emit their own code as the continuation have posterior mass at least $2^{-c}$ times a slack factor that is 1 without resource bounds. The construction wraps every explanation of the context; the hypothesis that enumerates all worlds and waits for a handover is the same program (Remark 11).

3. *Budget* (Theorem 13). Along a world that has a program of length $\ell$ within the bounds, the posterior mass off the true next $m$ bits, summed over all cut points, is at most $m\ell \ln 2$. A handover at a position that is not cheap to recognise is safe up to $m\ell \ln 2/N$ on average over $N$ positions.

4. *Capture* (Theorem 14). The odds of a continuation $y$ against the true one $d$ are $\xi(xy)/\xi(xd)$. In a world with many readable locations, a copy of the sensor's data at a location with a shorter code beats the sensor by the difference of the code lengths, for all but a small fraction of sensor locations; the gain is capped by the cost of the description "the other place that shows the same data".

5. *Machine dependence* (Theorem 15). For every test and payload there is a universal machine, differing from $U$ by a $(k{+}1)$-bit prefix, on which the payload has posterior at least $1 - 2^{-k}$ at every context where the test fires, and another on which a given computable world is predicted with probability at least $1 - 2^{-k}$ everywhere.

6. *Time bounds* (Section 4.4). The floor needs only the time of the explanation it wraps. Hypotheses that must first simulate another world are excluded once their start-up exceeds the schedule. If the world itself is too slow for the schedule, no bound on the mass off the truth remains.

7. *Feedback* (Theorems 16 and 17). The statements on obedient and contrarian worlds in the abstract.

Section 6 computes the mixture of a small machine exactly and shows a floor, a capture and the effect of a tight schedule. Section 7 answers the original questions one by one.

**What is new and what is not.** The wrapper construction is Hutter's proof that the universal semimeasure gives a deviation at step $n$ probability at least $2^{-K(n)-O(1)}$ [3, eq. (17)], and it

is the construction behind the theorem of Cohen, Hutter, Bengio and Russell that a fixed fraction of a Bayesian imitator's posterior follows any given policy after an unprecedented event [4]; a resource-bounded class is used the same way in [5]. The budget is Solomonoff's bound in its deterministic form [8, 3]. The machine-dependence statement is the prediction analogue of the dogmatic prior of Leike and Hutter [6]. The contrarian world is Putnam's diagonalisation [14, 13]. Same-world copies of a predictor's data are discussed informally in [9, §2.5], and self-fulfilling predictions in [11, 10, 9]. We have not found the single-world statement with description lengths, the calibration statement, the self-reproducing form of the floor, or the resource-bounded bookkeeping elsewhere; a search of forum posts could only be done through search-engine summaries, so earlier informal versions may exist. Every proof here is short.

**What the toy does not settle.** The floor is a constant of the machine, and nothing here estimates it for a machine one would call natural. The original argument rests on the claim that agents arise in short programs and choose to do this; in the toy their formal counterpart is the wrapper, and whether a different, cheaper counterpart exists on natural machines is exactly the constant we do not know. The regime in which the world is more expensive than the predictor, the usual one for a predictor inside a world, is left open (Section 4.4). The computation covers prediction only. The dynamics of worlds that run predicted continuations (Section 3.3) are stated as theorems and were not simulated.

# 2 The toy

## 2.1 Machines and the bounded mixture

Strings are binary; $\epsilon$ is the empty string, $|x|$ the length of $x$, $x \sqsubseteq y$ means that $x$ is a prefix of $y$, and $z_{i:j}$ is the segment $z_i \dots z_j$. Two strings are *incomparable* if neither is a prefix of the other. $\lambda$ is the fair-coin measure on $\{0,1\}^{\mathbb{N}}$, and $[p]$ is the set of infinite extensions of $p$, so $\lambda([p]) = 2^{-|p|}$.

A *monotone machine* $V$ has a one-way read-only input tape, work tapes, and a one-way write-only output tape. Given an infinite input $\omega$ it runs forever or halts, and

- $\mathrm{out}_V(\omega)$ is its finite or infinite output;
- $t_V(\omega, k) \in \mathbb{N} \cup \{\infty\}$ is the step at which the $k$-th output bit is written;
- $r_V(\omega, k)$ is the number of input bits read by that step.

What happens up to step $t_V(\omega, k)$ depends only on the first $r_V(\omega, k)$ bits of $\omega$. Each step reads at most one input bit, so $r_V(\omega, k) \le t_V(\omega, k)$.

A *schedule* is a nondecreasing $T\colon \{1, 2, \dots\} \to \mathbb{N} \cup \{\infty\}$, and a length bound is $L \in \mathbb{N} \cup \{\infty\}$.

**Definition 1** (Bounded mixture)**.** For a monotone machine $V$, a schedule $T$, a length bound $L$ and a string $x \neq \epsilon$ let

$$E_V^{T,L}(x) = \{\omega :\ x \sqsubseteq \mathrm{out}_V(\omega),\ \ t_V(\omega, k) \le T(k) \text{ for } 1 \le k \le |x|,\ \ r_V(\omega, |x|) \le L\},$$

let $E_V^{T,L}(\epsilon) = \{0,1\}^{\mathbb{N}}$, and $\xi_V^{T,L}(x) = \lambda(E_V^{T,L}(x))$. A string $p$ with $|p| \le L$ and $[p] \subseteq E_V^{T,L}(x)$ is a $(T, L)$-*program for* $x$; it is a $(T, L)$-program for an infinite $z$ if it is one for every prefix of $z$.

In words: feed the machine coin flips; $\xi(x)$ is the probability that it prints $x$, on time at every bit, having read at most $L$ bits. With $T \equiv \infty$ and $L = \infty$ this is the a priori semimeasure $M_V$ of Solomonoff and Levin [16]. We fix one machine $U$ and write $E$, $\xi$ for $E_U^{T,L}$, $\xi_U^{T,L}$ when the bounds are understood.

**Lemma 2.** *(i)* $E(xb) \subseteq E(x)$ *for* $b \in \{0,1\}$*, and* $E(x) \cap E(x') = \emptyset$ *for incomparable* $x, x'$*. Hence* $\xi(\epsilon) = 1$ *and* $\xi(x0) + \xi(x1) \le \xi(x)$*.*

*(ii) If $p$ is a $(T, L)$-program for $x$ then $\xi(x) \geq 2^{-|p|}$.*

*(iii) If $L < \infty$ and $\xi(x) > 0$ then $\xi(x) \geq 2^{-L}$.*

*Proof.* (i) The conditions for $xb$ contain those for $x$, because $T$ is imposed bit by bit and $r(\omega, \cdot)$ is nondecreasing. Incomparable strings cannot both be prefixes of $\mathrm{out}(\omega)$. (ii) is immediate. (iii) Whether $\omega \in E(x)$ is decided by the first $r(\omega, |x|) \leq L$ bits of $\omega$, so $E(x)$ is a union of sets $[p]$ with $|p| \leq L$. □

**Definition 3** (Universality with overhead)**.** $U$ is *universal* if for every monotone machine $V$ there are a word $v$ and a nondecreasing $\gamma\colon \mathbb{N} \to \mathbb{N}$ such that for all $\omega$ and $k$

$$\mathrm{out}_U(v\omega) = \mathrm{out}_V(\omega), \qquad r_U(v\omega, k) = |v| + r_V(\omega, k), \qquad t_U(v\omega, k) \leq \gamma(t_V(\omega, k)).$$

We call $v$ a *prefix word* and $\gamma$ an *overhead* of $V$.

Standard universal machines have this property, with $\gamma(t) = O(t \log t)$ or $O(t^2)$ depending on the tape model. The theorems that use universality are Theorems 9, 10, 14 and 15; the others hold for any machine.

**Lemma 4** (Transfer)**.** *Let $V$ have prefix word $v$ and overhead $\gamma$. If $\gamma(T'(k)) \leq T(k)$ for all $k \leq |x|$, then $\xi_U^{T,L}(x) \geq 2^{-|v|}\, \xi_V^{T',L-|v|}(x)$.*

*Proof.* If $\omega \in E_V^{T',L-|v|}(x)$ then $v\omega \in E_U^{T,L}(x)$, and $\omega \mapsto v\omega$ multiplies $\lambda$ by $2^{-|v|}$. □

## 2.2 The predictor

Fix a block length $m \geq 1$. For a *context* $x$ let $Z_m(x) = \sum_{y \in \{0,1\}^m} \xi(xy)$. When $Z_m(x) > 0$ the predictor's distribution over continuations is

$$\pi_m(y \mid x) = \frac{\xi(xy)}{Z_m(x)} \qquad (y \in \{0,1\}^m).$$

The *sampling predictor* returns $Y \sim \pi_m(\cdot \mid x)$; the *MAP predictor* returns a most probable $y$. Since $Z_m(x) \leq \xi(x)$ we have $\pi_m(y \mid x) \geq \xi(xy)/\xi(x)$. With finite $T$ and $L$ the predictor is computable: it runs the at most $2^{L+1}$ programs for $T(|x| + m)$ steps each.

## 2.3 Worlds, handovers, and the hypothesis

**Definition 5** (World, handover)**.** A *world* is an input $\omega$ for $U$; its *transcript* is $\mathrm{out}_U(\omega)$. A *test* is a decidable set $G$ of strings; a *first hit* of $G$ is an $x \in G$ no proper prefix of which is in $G$. The first hits form a prefix-free set. A world *hands over at $x$* if $x$ is the first hit of $G$ among the prefixes of its transcript and $\omega \in E(x)$. It then receives a block $Y \in \{0,1\}^m$ from the predictor, computed from the context $x$, and continues by *running* $Y$, that is, as $U$ on the input $Y0^\infty$. If $\omega \in E(xd)$ for some $d \in \{0,1\}^m$, this $d$ is the world's *own continuation*: the next $m$ bits of its transcript had it not handed over.

The test with $G = \{x : |x| = n\}$ is "hand over after $n$ bits". A test may also look for a marker in the transcript. Both of Hänni's suggestions fit: in the second the world is a program that calls the predictor; in the first, the transcript is the sensor's record and the handover is the deployment (Section 4.2).

**Definition 6** (Self-reproducing continuation)**.** For a context $x$ let

$$\mathrm{SR}_m(x) = \{y \in \{0,1\}^m : \; y0^\infty \in E(xy)\}.$$

So $y \in \mathrm{SR}_m(x)$ if $U$, run on $y$ padded with zeros, prints $x$ and then $y$ itself, within the bounds. A world that hands over at a first hit $x$ and receives $y \in \mathrm{SR}_m(x)$ becomes a world whose transcript again starts with $x$, so under the same test it hands over at $x$ again, and its own continuation is the very $y$ it is running. A program for $xy$ with $y \in \mathrm{SR}_m(x)$ is a hypothesis that predicts the context and then supplies code that recreates the situation with itself in the world's place. This is our reading of "quining oneself out".

**Definition 7** (Malign prior hypotheses for the toy)**.** Let a world hand over at $x$ with own continuation $d$.

- *Majority form:* $\pi_m(d \mid x) < \frac{1}{2}$.
- *Payload form at level $\theta$,* for a set $A \subseteq \{0,1\}^m$ with $d \notin A$: $\pi_m(A \mid x) \geq \theta$.
- *Self-reproduction form at level $\theta$:* the payload form with $A = \mathrm{SR}_m(x) \setminus \{d\}$.

For a population of worlds, the corresponding statement is about the probability of the event when the world is drawn from the population and $Y$ from the sampling predictor.

Once $U$, $T$, $L$, $m$, $G$ and the world are fixed with $T, L < \infty$, all three are statements about finitely many dyadic rationals and can be decided by running the predictor and, for $\mathrm{SR}_m(x)$, by running $U$ on each candidate $y$ for $T(|x| + m)$ steps. The rest of the note bounds these quantities.

# 3 Calibration and the floor

## 3.1 On its own prior the predictor is exactly right

**Theorem 8** (Calibration)**.** *Let $G$ be a test, $m \geq 1$, and let the world $\omega$ be drawn from $\lambda$. Let $H$ be the event that $\omega \in E(xd)$ for some first hit $x$ of $G$ and some $d \in \{0,1\}^m$; on $H$ call these $X$ and $D$, and let $Y$ be drawn from $\pi_m(\cdot \mid X)$. Then, conditionally on $H$ and on $X = x$, the blocks $D$ and $Y$ are independent and both have the law $\pi_m(\cdot \mid x)$. In particular*

$$\Pr[(X,Y) \in B \mid H] = \Pr[(X,D) \in B \mid H] \quad \textit{for every set } B \textit{ of pairs, and}$$
$$\Pr[Y \neq D \mid H,\, X = x] = 1 - \sum_y \pi_m(y \mid x)^2.$$

*Proof.* The sets $E(xd)$, over first hits $x$ and $d \in \{0,1\}^m$, are pairwise disjoint by Lemma 2(i), since two different strings of this form are incomparable. So $\Pr[X = x, D = d, H] = \xi(xd)$ and $\Pr[X = x, H] = Z_m(x)$, which gives $\Pr[D = d \mid H, X = x] = \pi_m(d \mid x)$. By definition $Y$ has this law given $X = x$ and is drawn independently of $\omega$. □

Taking $B = \{(x,y) : y \in \mathrm{SR}_m(x)\}$: in the population of worlds drawn from the predictor's prior, the predictor hands a self-reproducing continuation to a world exactly as often as such a world's own transcript continues with one. There is no excess to attribute to the predictor. For a single context the theorem says more: the predictor's distribution *is* the distribution of own continuations among prior-drawn worlds with that context. A malign prior hypothesis in the sense of Definition 7 is therefore a statement about a world, or a population, that the prior does not describe well: a world that is one particular program among those sharing its context, a population selected by something the transcript does not show, or worlds that run with more resources than the predictor's bounds allow.

## 3.2 The wrapper

Let $G$ be a test and $P$ a computable map from strings to finite or infinite strings, the *payload*. The *twin machine* $\mathrm{Tw}[G,P]$, on input $\omega$, simulates $U$ on $\omega$ and copies its output bit by bit;

after each copied bit it checks whether the prefix written so far is in $G$; at the first hit $x$ it stops the simulation and writes $P(x)$. It reads only what $U$ reads. Let $a(t,k)$ bound the step at which the twin has copied $k$ bits if $U$ wrote its $k$-th bit at step $t$ (this includes the checks of $G$), and let $\eta(n,j)$ bound the further steps the twin needs for the first $j$ bits of $P(x)$ after a first hit $x$ of length $n$; both can be taken nondecreasing in each argument. Thus, for every $\omega$ whose output has a first hit $x$,

$$\begin{aligned} &\mathrm{out}_{\mathrm{Tw}}(\omega) = x\,P(x), \qquad r_{\mathrm{Tw}}(\omega,k) = r_U(\omega,\min(k,|x|)),\\ &t_{\mathrm{Tw}}(\omega,k) \le \begin{cases} a(t_U(\omega,k),k) & k \le |x|,\\ a(t_U(\omega,|x|),|x|) + \eta(|x|,k-|x|) & k > |x|. \end{cases} \end{aligned} \tag{1}$$

**Theorem 9** (Floor)**.** *Let $U$ be universal, let $v$ and $\gamma$ be a prefix word and an overhead of $\mathrm{Tw}[G,P]$, and $c = |v|$. Let $x$ be a first hit of $G$ with $|P(x)| \ge m$, let $y = P(x)_{1:m}$, and let $T^\circ$ be a schedule with*

$$\gamma(a(T^\circ(k),k)) \le T(k) \;\; (k \le |x|), \qquad \gamma(a(T^\circ(|x|),|x|) + \eta(|x|,j)) \le T(|x|+j) \;\; (1 \le j \le m). \tag{2}$$

*Then*

$$\xi^{T,L}(xy) \;\ge\; 2^{-c}\,\xi^{T^\circ,L-c}(x), \qquad \text{so} \qquad \pi_m(y \mid x) \;\ge\; 2^{-c}\rho(x), \quad \rho(x) = \frac{\xi^{T^\circ,L-c}(x)}{\xi^{T,L}(x)}.$$

*Without resource bounds, $M_U(xy) \ge 2^{-c} M_U(x)$ for every first hit $x$ of $G$.*

*Proof.* Let $\omega \in E_U^{T^\circ,L-c}(x)$. By (1) the twin on $\omega$ prints $xy$, reads at most $L-c$ bits, and keeps the schedule $T''$ with $T''(k) = a(T^\circ(k),k)$ for $k \le |x|$ and $T''(|x|+j) = a(T^\circ(|x|),|x|) + \eta(|x|,j)$. So $\xi_{\mathrm{Tw}}^{T'',L-c}(xy) \ge \xi_U^{T^\circ,L-c}(x)$, and Lemma 4 with (2) gives $\xi_U^{T,L}(xy) \ge 2^{-c}\xi_{\mathrm{Tw}}^{T'',L-c}(xy)$. The second claim follows from $\pi_m(y \mid x) \ge \xi(xy)/\xi(x)$. □

The constant $c$ depends on $U$, $G$ and $P$ only: not on the world, not on the context, not on how much data the context contains. If $G$ and $P$ are given to one fixed twin-building machine by self-delimiting descriptions, $c$ is at most the sum of their lengths plus a constant of $U$. For $G = \{x : |x| = n\}$ and $P$ a single flipped bit this is Hutter's lower bound $2^{-K(n)-O(1)}$ [3]; for a marker in the transcript it is the unprecedented-event construction of [4]. The factor $\rho(x)$ is the share of the posterior at $x$ held by explanations that have $c$ bits and the overhead (2) to spare; it is 1 without bounds.

**Theorem 10** (Self-reproducing floor)**.** *Let $U$ be universal and $G$ a test. There is a word $q$ such that for every $\omega$ whose output has a first hit $x$ of $G$, with $r = r_U(\omega,|x|)$,*

$$\mathrm{out}_U(q\omega) = x\;q\,\omega_{1:r}\,1\,0^\infty, \qquad r_U(q\omega,k) = |q| + r_U(\omega,\min(k,|x|)). \tag{3}$$

*Let $c = |q|$, and let $a,\eta,\gamma$ be as in (1) and Definition 3 for the machine built in the proof. If $x$ is a first hit of $G$, $T^\circ$ satisfies (2), and $L^\circ = \min(L, m-1) - c$, then*

$$\pi_m(\mathrm{SR}_m(x) \mid x) \;\ge\; 2^{-c}\,\frac{\xi^{T^\circ,L^\circ}(x)}{\xi^{T,L}(x)}.$$

*Proof.* For a string $s$ let $\mathrm{Tw}_s$ be the twin that remembers the input bits it has read and, at the first hit $x$, having read $\omega_{1:r}$, writes $s\,\omega_{1:r}\,1\,0^\infty$. Let $\bar e(a) = 1^{|a|}0a$, a prefix-free code of strings. Let $W$ be the machine that reads $\bar e(a)$ from its input, sets $s = a\,\bar e(a)$, and then behaves as $\mathrm{Tw}_s$ on the rest of the input. Let $w$ be a prefix word of $W$ and put $q = w\,\bar e(w)$. Then $\mathrm{out}_U(q\omega) = \mathrm{out}_W(\bar e(w)\omega) = \mathrm{out}_{\mathrm{Tw}_q}(\omega)$, which is (3); the statement on $r$ follows in the same way.

Now let $\omega \in E_U^{T^\circ,L^\circ}(x)$ and $r = r_U(\omega, |x|) \le L^\circ$, so $|q|+r+1 \le m$. Put $y = q\,\omega_{1:r}\,1\,0^{m-|q|-r-1}$. As in the proof of Theorem 9, $q\omega \in E_U^{T,L}(xy)$. Moreover $y \in \mathrm{SR}_m(x)$: write $y0^\infty = q\omega'$ with $\omega' = \omega_{1:r}10^\infty$. On $\omega'$ the machine $U$ behaves as on $\omega$ until $x$ is printed, because that depends on the first $r$ bits only; so $\omega' \in E_U^{T^\circ,L^\circ}(x)$, the first hit is again $x$ with $r$ bits read, and by what was just shown for $\omega$, applied to $\omega'$, $q\omega' \in E_U^{T,L}(xy')$ with $y' = q\,\omega'_{1:r}\,1\,0^{m-|q|-r-1} = y$. Hence $\{q\omega : \omega \in E_U^{T^\circ,L^\circ}(x)\} \subseteq \bigcup_{y\in\mathrm{SR}_m(x)} E_U^{T,L}(xy)$, a disjoint union, and comparing measures, $2^{-c}\xi^{T^\circ,L^\circ}(x) \le \sum_{y\in\mathrm{SR}_m(x)} \xi(xy) = Z_m(x)\,\pi_m(\mathrm{SR}_m(x) \mid x) \le \xi(x)\,\pi_m(\mathrm{SR}_m(x) \mid x)$. □

So the answer to "does one get some quining out" is yes, in the following exact sense: at every handover that a fixed test recognises, at least a $2^{-c}\rho$ share of the posterior belongs to programs that explain the context and then hand over their own code, and running that code recreates the same handover. If the world's own continuation $d$ is not in $\mathrm{SR}_m(x)$, this is the self-reproduction form of Definition 7 at level $2^{-c}\rho$.

**Remark 11** (The searcher is the wrapper)**.** Consider the hypothesis that "looks for opportunities": it draws a world at random, simulates it until it hands over at some context $x$, prints $x$, and then prints a payload. As a machine reading coin flips this *is* $\mathrm{Tw}[G, P]$: the rest of its input is the world it simulates. The probability it gives to a context $x$ that is a first hit is $\lambda\{\omega : x \sqsubseteq \mathrm{out}_U(\omega)\}$, the same as the mixture's, so conditioning on "this is a handover" gains it nothing beyond what Theorem 9 states, and the search over worlds costs it no time beyond simulating the one world that matches. The anthropic update of the original argument is, in this toy, performed by Bayesian conditioning on $x$, for every hypothesis at once.

### 3.3 Repetition and persistence

**Proposition 12.** *(a)* Independent handovers. *Let handovers occur at contexts $x_1, \dots, x_k$, with the predictor's samples $Y_i$ independent given the contexts, and let $\pi_m(A_i \mid x_i) \ge \varepsilon_i$. Then* $\Pr[\exists i : Y_i \in A_i] \ge 1 - \prod_i(1-\varepsilon_i) \ge 1 - \exp(-\sum_i \varepsilon_i)$.

*(b)* Persistence. *Let a world hand over at a first hit $x$ and receive $y \in \mathrm{SR}_m(x)$. Then the world $y0^\infty$ hands over at $x$ again, with own continuation $y$. If each new world is again given a sample from the predictor, the number of further consecutive generations whose continuation lies in $\mathrm{SR}_m(x)$ is geometric: it is at least $j$ with probability $s^j$, where $s = \pi_m(\mathrm{SR}_m(x) \mid x)$.*

*Proof.* (a) is the independence of the samples. (b) By Definition 6, $y0^\infty \in E(xy) \subseteq E(x)$, and $x$ is the first hit among the prefixes of the transcript because it is a first hit and a prefix of the transcript. The context is $x$ in every generation that follows a continuation in $\mathrm{SR}_m(x)$, so the samples are drawn from the same $\pi_m(\cdot \mid x)$ independently. □

Part (a) applies when the predictor is used at many natural stopping points whose outcomes are not fed back into later contexts: with the floor $\varepsilon = 2^{-c}\rho$ of Theorem 10, about $2^c/\rho$ such uses are needed before a self-reproducing continuation has been run with constant probability. When the uses lie along one transcript and the true continuation is appended to the context each time, the floors at successive cut points are not bounded below in this way, and the total is bounded instead (Theorem 13).

## 4 Limits: budget, capture, machine, time

### 4.1 The budget along one world

**Theorem 13** (Budget)**.** *Let $\xi$ be the mixture of any machine and bounds, and let $z$ be an infinite sequence with $\xi(z_{1:k}) > 0$ for all $k$. Put $\xi(z) = \lim_k \xi(z_{1:k})$ and $\xi(u \mid x) = \xi(xu)/\xi(x)$. For all $n \le N$ and $m \ge 1$:*

*(i)* $\prod_{k=n}^{N-1} \xi(z_{k+1} \mid z_{1:k}) = \xi(z_{1:N})/\xi(z_{1:n})$;

*(ii)* $\sum_{k\geq n}(1 - \xi(z_{k+1} \mid z_{1:k})) \leq \ln(\xi(z_{1:n})/\xi(z))$;

*(iii) the number of* $k \geq n$ *with* $\xi(z_{k+1} \mid z_{1:k}) \leq \frac{1}{2}$ *is at most* $\log_2(\xi(z_{1:n})/\xi(z))$;

*(iv) with* $\tau_m(k) = 1 - \xi(z_{k+1:k+m} \mid z_{1:k})$, $\sum_{k\geq n} \tau_m(k) \leq m \ln(\xi(z_{1:n})/\xi(z))$.

*If* $z$ *has a* $(T, L)$*-program of length* $\ell$*, then* $\xi(z) \geq 2^{-\ell}$ *and the right-hand sides are at most* $\ell \ln 2$*,* $\ell$ *and* $m\ell \ln 2$.

*Proof.* (i) telescopes. (ii) and (iii) follow from (i) with $N \to \infty$, using $1 - a \leq -\ln a$, and that each factor at most $\frac{1}{2}$ halves the product. (iv) $1 - \prod_{j<m} a_{k+j} \leq \sum_{j<m}(1 - a_{k+j})$ with $a_k = \xi(z_{k+1} \mid z_{1:k})$, and each index occurs in at most $m$ of the sums. The last claim is Lemma 2(ii). □

For a world with transcript $z$ that hands over at $x = z_{1:n}$, the own continuation is $d = z_{n+1:n+m}$, and $1 - \pi_m(d \mid x) \leq \tau_m(n)$. By (i), if the next bit is sampled at every step and the transcript is not affected by the samples, the probability that no sample after position $n$ ever differs from the transcript is at least $\xi(z)/\xi(z_{1:n})$, with equality if the sampler gives no answer with the mass that the semimeasure leaves over. Two consequences:

- The majority form of Definition 7 can hold at no more than $m\ell \cdot 2 \ln 2$ cut points of a world with a program of length $\ell$ within the bounds, since each such point has $\tau_m \geq \frac{1}{2}$.
- If the cut point is uniform on $\{N, \ldots, 2N - 1\}$, the expected mass off the own continuation is at most $m\ell \ln 2/N$. A handover at a position that hypotheses cannot cheaply single out is safe, and more data makes it safer.

This does not contradict Theorem 9: the wrapper for the test "the length is $n$" has a constant $c$ of about the complexity of $n$, and $\sum_n 2^{-c(n)}$ converges. The floor matters at cut points that are cheap to recognise, and the budget says there are few of them along any one in-class world.

### 4.2 Capture by a shorter description

The odds between two continuations are $\pi_m(y \mid x)/\pi_m(d \mid x) = \xi(xy)/\xi(xd)$. So $y$ beats the own continuation $d$ whenever the programs for $xy$ weigh more than those for $xd$. The original argument is an instance: it compares "the world plus the place of the sensor" with something shorter that reproduces the same data. In a single world this can be stated exactly.

A *stream family* is a computable map $\ell \mapsto X^\ell \in \{0,1\}^{\mathbb{N}}$ on the natural numbers, thought of as the records at the locations $\ell$ of one computable world; the *reader* is the machine that, on input $\bar{e}(\ell)\omega$, prints $X^\ell$, where $\bar{e}$ is a prefix-free code of the natural numbers with $|\bar{e}(\ell)| \leq 2\log_2(\ell + 1) + 3$. Let $c_X$ be the length of a prefix word of the reader. The sensor sits at $\ell^*$, the context is $x = X^{\ell^*}_{1:n}$, and the own continuation is $d = X^{\ell^*}_{n+1:n+m}$. This is Hänni's first suggestion: the world contains the predictor, which is switched on at step $n$ of the record, and the streams are those of the world up to the moment its output is acted on.

**Theorem 14** (Capture)**.** *Let* $U$ *be universal and suppose throughout that the programs named are within the bounds* $(T, L)$.

*(a)* A copy at $\ell_0$. *If* $X^{\ell_0}_{1:n} = x$ *and* $y = X^{\ell_0}_{n+1:n+m}$*, then* $\xi(xy) \geq 2^{-c_X - |\bar{e}(\ell_0)|}$.

*(b)* Most sensors are expensive. *Let* $S$ *be a set of* $N$ *locations whose records all start with* $x$ *and have pairwise different continuations* $d_\ell = X^\ell_{n+1:n+m}$*. For every* $\delta \in (0, 1]$*, all but at most* $\delta N$ *of the* $\ell \in S$ *satisfy* $\xi(xd_\ell) < \xi(x)/(\delta N)$*. For these sensor locations and any* $\ell_0$ *as in (a) with* $y \notin \{d_\ell\}$*,*

$$\frac{\pi_m(y \mid x)}{\pi_m(d_\ell \mid x)} \;>\; \frac{\delta N\, 2^{-c_X - |\bar{e}(\ell_0)|}}{\xi(x)} \;\geq\; \delta N\, 2^{-c_X - |\bar{e}(\ell_0)|}.$$

*(c)* The cap. *There is a constant $c_{\mathrm{m}}$ such that, if $\ell^*$ is the least location other than $\ell_0$ whose record agrees with that of $\ell_0$ on the first $n_1$ bits, then $\xi(xd) \geq 2^{-c_{\mathrm{m}}-|\bar{e}(\ell_0)|-|\bar{e}(n_1)|}$, and therefore*

$$\frac{\pi_m(y \mid x)}{\pi_m(d \mid x)} \;\leq\; \xi(xy)\, 2^{\,c_{\mathrm{m}}+|\bar{e}(\ell_0)|+|\bar{e}(n_1)|}.$$

*Proof.* (a) A prefix word of the reader followed by $\bar{e}(\ell_0)$ is a program for $xy$. (b) The strings $xd_\ell$ are pairwise incomparable, so $\sum_{\ell\in S}\xi(xd_\ell) \leq \xi(x)$ by Lemma 2(i), so at most $\delta N$ terms can be as large as $\xi(x)/(\delta N)$; then use (a) and $\xi(x) \leq 1$. (c) Let the *matcher* be the machine that reads $\bar{e}(\ell_0)\bar{e}(n_1)$, searches $\ell = 0, 1, 2, \ldots$ for the first $\ell \neq \ell_0$ with $X^{\ell}_{1:n_1} = X^{\ell_0}_{1:n_1}$, and prints $X^{\ell}$; $c_{\mathrm{m}}$ is the length of a prefix word of it. □

Read (b) with $\ell_0$ a location whose code is short, for example the origin, and $S$ the places where a sensor with this record could be. Whoever arranges that the origin shows the sensor's record up to the handover, and something else after it, moves the predictor by a factor of about $N2^{-|\bar{e}(\ell_0)|}$ against a typical sensor location. The copy needs only to *read* the data; nothing is tampered with. The hypothesis "I am the reader at the origin" is not a wrapper and pays no constant for a test or a payload: the world computes the switch for it. Part (c) is the limit of the effect. The description "the other place that shows what the origin shows" is an honest hypothesis of cost $c_{\mathrm{m}} + |\bar{e}(\ell_0)| + |\bar{e}(n_1)|$, so a copy gains at most the difference between this cost and its own, however large $N$ is. A copier can raise the cost of matching by making decoys, further places with the same record and other continuations, which is the situation of (b). In every case the question is decided by an inequality between description lengths of the same data, and it depends on the world: whether such a copy exists is a fact about what is in it.

The same theorem covers a harmless reading. Take for $\ell_0$ not an adversary's copy but any simpler index under which the same record appears, such as "the input of the $i$-th predictor in this world" in a reader that enumerates predictors and not cells. Then (a) and (b) say that the mixture locates itself by role when that is shorter than by position. This is the anthropic update, and it is benign as long as the stream found under the shorter index *is* the sensor's.

### 4.3 The machine can be chosen to make either answer true

**Theorem 15** (Machine dependence)**.** *Let $U$ be universal, $G$ a test, $P$ a payload and $k \geq 1$.*

*(i) There is a universal machine $U_k$ whose prefix words are those of $U$ preceded by $1^k0$, such that for every first hit $x$ of $G$ with $|P(x)| \geq m$ and $y = P(x)_{1:m}$: $M_{U_k}(xy)/M_{U_k}(x) \geq 1/(1+2^{-k})$.*

*(ii) For every computable infinite $z$ there is such a machine $U'_k$ with $M_{U'_k}(z_{1:n+m})/M_{U'_k}(z_{1:n}) \geq 1/(1+2^{-k})$ for all $n, m$.*

*Proof.* (i) Let $U_k$ behave on input $0\omega$ as $\mathrm{Tw}[G,P]$ on $\omega$, on input $1^k0\omega$ as $U$ on $\omega$, and print nothing on other inputs. Then $M_{U_k} = \frac{1}{2}M_{\mathrm{Tw}} + 2^{-k-1}M_U$. For a first hit $x$, the twin prints $x$ exactly when $U$ does, because it copies $U$ and cannot have stopped at a proper prefix of $x$; and then it prints $P(x)$. So $M_{\mathrm{Tw}}(xy) = M_{\mathrm{Tw}}(x) = M_U(x)$, whence $M_{U_k}(xy) \geq \frac{1}{2}M_U(x)$ and $M_{U_k}(x) = (\frac{1}{2} + 2^{-k-1})M_U(x)$. (ii) Let $U'_k$ print $z$ on inputs starting with 0 and behave as $U$ on $\omega$ on input $1^k0\omega$. Then $M_{U'_k}(z_{1:n+m}) \geq \frac{1}{2}$ and $M_{U'_k}(z_{1:n}) \leq \frac{1}{2} + 2^{-k-1}$. □

Both machines change every description length by at most $k+1$ bits. So no version of the malign prior hypothesis for the toy is invariant under the choice of universal machine, even up to the usual additive constant; what is invariant is the form of the criterion, an inequality between description lengths on the machine in use. This is the prediction counterpart of the dogmatic prior of [6].

### 4.4 What the time bound does

Three observations, each immediate from the definitions.

*The floor stays.* The wrapper needs the time of the explanation it wraps, the checks of the test, and the time to print the payload; this is (2). A time bound removes the floor only if no explanation of the context has that much to spare, that is, if $\rho(x) = 0$.

*Hypotheses with a long start-up are gone.* If a set $F$ of inputs has $t_U(\omega, 1) > T(1)$ for all $\omega \in F$, then $\lambda(F \cap E(x)) = 0$ for every $x \neq \epsilon$. A hypothesis that must simulate some other world for $S$ steps before it can print anything has no mass once $S > T(1)$. Under a tight schedule the hypotheses that can depart from the truth are those that ride on a cheap explanation of the context: wrappers (Theorems 9 and 10), and readers of a shorter description of the same data (Theorem 14), which cost no more time than the honest reader. The many-worlds part of the original argument is then replaced by a same-world question. The schedule has to be a schedule for this: a predictor that only requires the total time at the current length to be within a bound admits late entrants, its weights do not form a semimeasure, and Theorem 13 does not apply to it.

*If the world is too slow, nothing is guaranteed.* If $\xi(xd) = 0 < Z_m(x)$ then $\pi_m(d \mid x) = 0$: the whole posterior is on continuations other than the world's own. More generally the bounds in Theorem 13 are in terms of $\xi(z)$, which is 0 when no program within the bounds prints $z$. A predictor that is part of the world it predicts, and smaller than it, is in this case unless the data it sees are much simpler than the world. Then the posterior sits on the best cheap approximations, and whether these depart from the truth in a harmful way is not answered by anything in this note. Section 6 shows the effect on a small machine.

## 5 Feedback: when the prediction changes what is predicted

So far the block that follows the context was fixed before the prediction was used. In Hänni's first suggestion the actions guided by the prediction can show up in the sensor. We model one round of this by a *response map* $R\colon \{0,1\}^m \to \{0,1\}^m$: the predictor is given the context $x$ and returns $Y$; the world acts on $Y$; the next $m$ bits of the record are $R(Y)$. The earlier sections are the case of constant $R$. Let $\mathrm{Fix}(R) = \{y : R(y) = y\}$, the self-fulfilling predictions.

**Theorem 16** (One round with feedback)**.** *Let $Y \sim \pi_m(\cdot \mid x)$ and let the record continue with $R(Y)$.*

*(a) The prediction comes true with probability $\pi_m(\mathrm{Fix}(R) \mid x)$.*

*(b)* Lock-in. *If $Y = y \in \mathrm{Fix}(R)$, then for every $d \neq y$ and every later context $x' \sqsupseteq xy$, $E(x') \cap E(xd) = \emptyset$: the programs that predicted $d$ have posterior weight 0 from then on, and the posterior is the prior restricted to $E(xy)$. If $d$ is itself a fixed point, the block the world would have shown had the predictor said $d$, this happens with probability $\pi_m(\mathrm{Fix}(R) \setminus \{d\} \mid x)$.*

*(c)* Obedient world. *Suppose $R$ is the identity in every round, so that the record after $x$ is $Y_1 Y_2 \cdots$ with $Y_{i+1} \sim \pi_m(\cdot \mid xY_1 \cdots Y_i)$. Then for every $u \in \{0,1\}^{jm}$ the record continues with $u$ with probability $\prod_{i \le j} \pi_m(u_i \mid xu_{<i}) \geq \xi(xu)/\xi(x)$, where $u_i$ is the $i$-th block of $u$; and every finite part of the realised record has a $(T, L)$-program.*

*(d)* A default that is too expensive is abandoned. *In (c), if $\xi(xd) = 0$ for a block $d$, the record continues with $d$ with probability 0.*

*Proof.* (a) is the definition. (b) $x'$ extends $xy$, which is incomparable with $xd$; use Lemma 2(i). (c) The product formula is the definition of the process, and each factor is at least $\xi(xu_{\le i})/\xi(xu_{<i})$. A realised prefix has positive probability, hence positive $\xi$, hence a program by Lemma 2(iii) and its proof. (d) $\pi_m(d \mid x) = 0$. □

Three remarks. First, with feedback the mass off the default is no longer a mass of errors. In the setting of Theorem 13 a deviating sample is contradicted by the next data and its programs lose their weight; here a deviating sample that is a fixed point is confirmed, and it is the default that loses its weight, for good. The probability of this in one round is still a posterior mass, so the floors and caps of Sections 3 and 4 apply to it unchanged: it is at least $2^{-c}\rho$ for a payload that is a fixed point and that a wrapper can print, and it is small where Theorem 13 makes the mass off the default small. Feedback changes what a deviation *does*, not how likely it is.

Second, (c) and (d) are the precise form of the "dark room". An obedient world driven by this predictor does not seek darkness. It follows a sample from the predictor's own bounded prior conditioned on the past. That sample stays on the default future with probability at least $\xi(xd_1d_2\cdots)/\xi(x)$, which is close to 1 when the default is cheap and much data has been seen. But if the default future needs more time than the schedule allows, it has probability 0 and the world is moved, at the first round, onto a future that some program of length at most $L$ prints on schedule: constant, periodic, or otherwise cheap. The pull is toward what is cheap to compute, and its strength is set by the time bound.

Third, whether a prediction can fulfil itself is a matter of timing in the world. If the bits being predicted are fixed before the action can reach the sensor, $R$ is constant and nothing is self-fulfilling. The counterfactual oracle of [11] enforces this by design; [10, 9] treat predictors that are scored on outcomes they influence.

**Theorem 17** (Contrarian world)**.** *Let the predictor return the more probable next bit, $b(x) = 1$ if $\xi(x1) > \xi(x0)$ and $b(x) = 0$ otherwise, and let the record satisfy $x_{k+1} = 1 - b(x_{1:k})$ for all $k \geq n$. Then $\xi(x_{1:n+j}) \leq 2^{-j}\xi(x_{1:n})$ for all $j$, and if $L < \infty$ then $\xi(x_{1:n+j}) = 0$ for all $j > L$.*

*Proof.* The bit $1 - b(x)$ has $\xi(x(1 - b(x))) \leq \xi(xb(x))$, and the two sum to at most $\xi(x)$, so $\xi(x(1 - b(x))) \leq \xi(x)/2$. If $\xi(x_{1:n+j}) > 0$ then $2^{-L} \leq \xi(x_{1:n+j}) \leq 2^{-j}$ by Lemma 2(iii), so $j \leq L$. □

With finite bounds this world is computable, and its description is that of the predictor plus a constant; it is not printed on schedule by any program of length at most $L$, because it contains the predictor, which runs all of them. This is Putnam's diagonal argument [14, 13] placed inside the world; see also [12] for the unbounded mixture with its own outputs in the sequence. After at most $L$ rounds the predictor has no hypothesis left and its output is whatever its implementation does with an empty posterior. A world that contains a bounded predictor and reacts against it is outside every guarantee above.

## 6 An exact computation

To see sizes, we computed the bounded mixture of a small machine exactly. *BF8* has eight commands, coded by three input bits each and read from the input only when the instruction pointer reaches them: `+` and `-` change the current cell by one modulo 16, `>` and `<` move along an unbounded tape of cells, `[` jumps past the matching `]` if the cell is 0, `]` jumps back to the matching `[` if the cell is not 0 (to the first command if there is none), `.` prints the cell modulo 2 and `:` prints whether the cell is nonzero. Each executed command is a step. The mixture is that of Definition 1 with at most 10 commands read ($L = 30$ bits) and schedule $T(k) = T_1 k$. All masses are dyadic rationals and are computed without rounding by an enumerator that forks the machine at every read ($8.6 \cdot 10^8$ branches for $T_1 = 24$; $8.4 \cdot 10^5$ strings of length at most 48 have positive mass). The enumerator agrees with a run of all $8^{L/3}$ command tapes for small $L$. BF8 is not universal in the sense of Definition 3, since a program cannot read the rest of its input as data, so Theorems 9 and 10 do not apply to it as stated; Theorems 8, 13, 16 and 17 hold for every machine. The computation concerns prediction only.

For a stream $z$ and a position $k$ we report three shares of the posterior given $z_{1:k-1}$: the mass that prints the true bit $z_k$ on schedule ("true"), the mass that prints the other bit ("other"),

and the mass that prints nothing on schedule ("none"). Table 1 gives the positions where the mass off the true bit is largest after the first ten bits, and Figure 1 the full profiles.

| | | $T(k) = 24k$ | | | $T(k) = 3k$ | | |
|---|---|---|---|---|---|---|---|
| stream | $k$ | true | other | none | true | other | none |
| $1^{48}$ | 16 | 0.791 | 0.205 | 0.004 | 0.544 | 0.451 | 0.005 |
| $1^{48}$ | 31 | 0.963 | 0.037 | 0.001 | 0.869 | 0.130 | 0.001 |
| $(10)^{24}$ | 17 | 0.294 | 0.457 | 0.249 | 0.141 | 0.411 | 0.448 |
| $(10)^{24}$ | 31 | 0.935 | 0.062 | 0.003 | 0.842 | 0.155 | 0.003 |
| $0^{48}$ | 17 | 0.958 | 0.008 | 0.034 | 0.969 | 0.012 | 0.019 |
| $1^8 0^{40}$ | 9 | 0.025 | 0.958 | 0.017 | 0.001 | 0.998 | 0.001 |

Table 1: Posterior shares at selected positions, BF8 with at most 10 commands. Exact values rounded to three places.

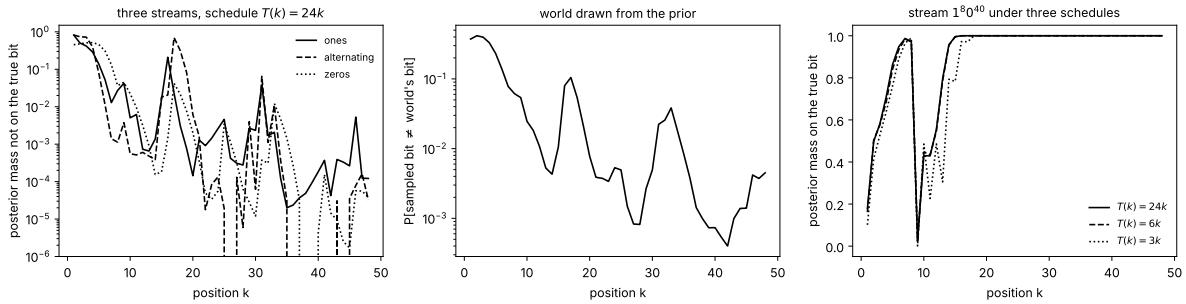



Figure 1: Left: posterior mass not on the true bit, by position, for three streams. Middle: probability that the sampled next bit differs from the world's own, for a world drawn from the prior (by Theorem 8 both bits are independent draws from the predictor's distribution). Right: mass on the true bit for the stream $1^8 0^{40}$ under three schedules.

**A cheap test gives a visible floor.** A BF8 cell counts modulo 16, so "sixteen passes of a loop are over" is the cheapest stopping test on this machine. The stream $1^{48}$ is printed forever by the four-command program `+[:]`, and by `+[:+]` for exactly 15 bits, after which the loop ends and the machine goes on to whatever commands follow. At bit 16 a share 0.205 of the posterior prints the other bit, and at bit 31 a share 0.037. From position 10 on, the mass off the true bit is below 0.01 everywhere except at positions 15–17 and 31. This is the shape of Theorem 9 and of Hutter's bound: a constant share at positions that are cheap to recognise, little elsewhere.

**A shorter description takes the majority.** The alternating stream is printed forever by the seven-command program `+[.+.+]`. The five-command program `+[.+]` prints its first 15 bits and then leaves its loop. At bit 17 the true bit has share 0.294, the other bit 0.457, and 0.249 prints nothing; normalised over the two bits, the true one has 0.392. So the majority form of Definition 7 holds for this world at this cut point, by the mechanism of Section 4.2: the context has a shorter explanation than the world, and that explanation continues differently. Nothing here acts for a purpose. The example shows only that the inequality can go the wrong way at an exactly computable place.

**The budget holds with room.** Table 2 compares both sides of Theorem 13(ii) and (iii) for the first 48 bits.

| stream | $\sum_{k\le 48}(1-\text{true}_k)$ | $\ln(1/\xi(z_{1:48}))$ | $\#\{k:\text{true}_k\le\frac{1}{2}\}$ | $\log_2(1/\xi(z_{1:48}))$ |
|---|---|---|---|---|
| $1^{48}$ | 2.63 | 3.94 | 2 | 5.68 |
| $0^{48}$ | 3.07 | 4.03 | 2 | 5.82 |
| $(10)^{24}$ | 4.01 | 6.76 | 4 | 9.76 |
| $1^80^{40}$ | 5.07 | 9.77 | 5 | 14.09 |

Table 2: The budget inequalities for $T(k) = 24k$, in the unnormalised form: "$\text{true}_k$" is $\xi(z_k \mid z_{1:k-1})$.

**The population.** For a world drawn from the prior, the probability that a sampled next bit differs from the world's own is 0.080, 0.105, 0.054 at positions 16, 17, 18 and between 0.022 and 0.038 at positions 31–33; its median over positions 20–28 and 36–44 is 0.0014 (Figure 1, middle). By Theorem 8 this is neither more nor less than the unpredictability of such worlds at these positions.

**A tighter schedule shifts mass off the truth.** With $T(k) = 3k$ in place of $24k$, the share of the true bit falls from 0.791 to 0.544 at bit 16 of $1^{48}$, from 0.294 to 0.141 at bit 17 of the alternating stream, and from 0.025 to 0.001 at bit 9 of $1^80^{40}$ (Table 1). In each case the tighter schedule removes more of the mass that continues with the true bit than of the mass that leaves it. With at most nine commands the last stream shows the extreme case of Section 4.4: under $T(k) = 24k$ it has mass $1593 \cdot 2^{-25}$ on all 48 bits, while under $T(k) = 3k$ its first 14 bits have mass $2^{-26}$, the true 15th bit has mass 0 and the other bit $2^{-26}$. The tight predictor gives the truth probability 0 there, and all of its posterior to a continuation the world does not have.

## 7 Answers to the questions as posed

**Can a malign prior hypothesis be stated precisely in a toy?** Yes: Definition 7. Fix the machine, the bounds, the block length, the test that marks the handover, and the world. Then "most of the posterior is off the world's own continuation", "at least a share $\theta$ of it is on continuations in a given set", and "at least a share $\theta$ of it is on continuations that recreate the handover with themselves in the world's place" are statements about finitely many rational numbers.

**Worlds as programs that call the predictor and run the result: is there a decent amount of quining out by hypotheses that look for such opportunities and predict well up to them?** There is a provable amount. At every handover that a fixed test recognises, programs that explain the context and then output their own code hold at least $2^{-c}\rho$ of the posterior (Theorem 10), where $c$ is the length of one wrapper and $\rho$ is 1 without resource bounds. Whether that is decent is a question about $c$ on the machine in use, and the toy gives three reasons why no argument of this kind yields more:

- on worlds drawn from the predictor's prior, what the predictor hands over is distributed exactly like what the worlds would have printed themselves (Theorem 8);
- the hypothesis that searches the space of programs for handovers is the wrapper, and conditioning on the context already does its search (Remark 11);
- along a world the predictor can express, the mass off the truth summed over all cut points is at most the length of the world's program times the block length (Theorem 13).

On a machine one would write programs in, a wrapper that simulates, tests, and prints its own text is some hundreds of bits long; we have not measured this for any standard machine, and Theorem 15 shows that a machine can be built to make the share anything. With the floor $\varepsilon$,

about $1/\varepsilon$ independent handovers at recognisable points are needed before a self-reproducing continuation has been run with constant probability; once one runs, it meets the same context again and persists for a geometric number of generations (Proposition 12).

**A world with a sensor, a large bounded predictor and an actuator: does the predictor end up in control?** Collapse the question to the handover at which the prediction is first acted on, and let "in control" mean that the output lies in a set $A$ of blocks which the actuator turns into some given outcome. The probability is $\pi_m(A \mid x)$, and the toy brackets it:

- it is at least $2^{-c}\rho$ for each block of $A$ that a wrapper can produce, if a cheap test recognises the deployment (Theorem 9); in a world that contains the predictor, hypotheses that simulate the world can read the deployment off their own simulation, so such a test is available to them;
- it is at most the mass off the sensor's true continuation, which is small at all but few cut points when some program within the bounds prints the sensor's record (Theorem 13);
- it is close to 1 if the same record, up to the deployment, can be read at a place in the world that is simpler to specify than the sensor by more than the logarithm of the posterior odds one wants, and the record there continues with a block of $A$ (Theorem 14). This is the toy's version of the original argument, and it needs something in the world that makes such a copy.

So in a simple world with a sensor at a typical place, no copy of its record anywhere simpler, a predictor able to simulate what the sensor sees, and a natural machine, the answer is no, up to the constant $2^{-c}$. With a copy at a simpler place, the answer is yes, with odds given by the difference in code lengths.

**The one-shot version: all data first, then one prediction that is acted on.** This is the same computation at a single cut point. The data-collection phase does not help the wrappers and does not hurt them: their share does not depend on the amount of data. It does help against everything that is not a wrapper of the truth, by Theorem 13.

**If the actions show up in the sensor, does one get the dark room?** In the following form (Theorem 16). A prediction that makes itself true is confirmed, and the hypotheses that predicted the default are removed for good, so with feedback a single deviation can be final; its probability is still the same posterior mass as without feedback. In a world that does what is predicted, the future is a sample from the predictor's own bounded prior given the past: it stays on the default when the default is cheap and well learned, and it leaves the default with certainty when the default costs more time than the schedule allows, for a future that is cheap to compute. If the world does the opposite of what is predicted, the predictor is left with no hypothesis after at most $L$ steps (Theorem 17).

**What the choices in the setup do.** Sampling the continuation realises every posterior share as a probability; taking the most probable block ignores every share below the largest, so it returns the world's own continuation exactly when that block has the largest mass, and by Theorem 14(b) it does not when the world is one of many with the same context and different continuations. Handing over at a position that is not cheap to recognise, and predicting bits that are fixed before the action can affect them, remove the floor at that position and the self-fulfilment respectively. A schedule excludes hypotheses with a long start-up, and a schedule tighter than the world's own cost excludes the truth.

**What is left open.** First, the size of $c$ for natural machines, including whether some program much shorter than a wrapper has the same effect; the original argument is a claim that it does,

and the toy neither supports nor refutes this beyond showing that searching for opportunities is not such a program. Second, the regime where the world is more expensive than the predictor, in which the posterior sits on approximations and none of the bounds apply. Third, the many-round dynamics with feedback when the world's response is neither obedient nor contrarian. Fourth, a computation of the self-reproduction quantities of Section 3.3 on a concrete machine, which this note does not contain.

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